

257

1006

Roll No.

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EVEN END SEMESTER EXAMINATION- 2024

Name of the Course: B. Tech
Name of the Paper: Finite Automata and Formal Languages

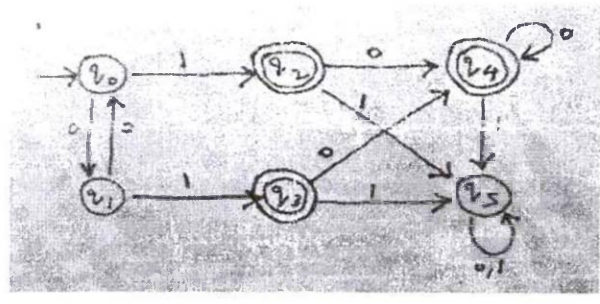
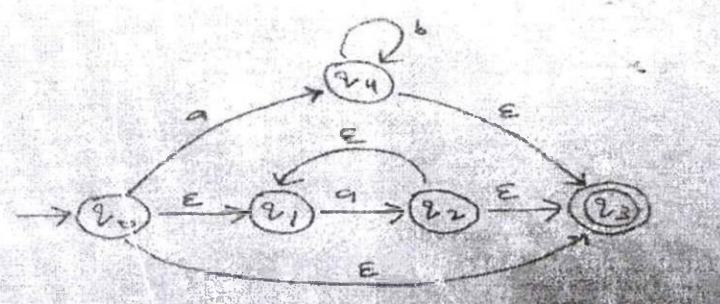
Semester: IV
Paper Code: TCS-402

Time: 3 Hours

Maximum Marks:100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20 marks)	
(a)	Design a Finite Automata over the alphabet $\Sigma = \{0, 1\}$ that accepts all the strings from the given language: $L = \{0^m 1^n \mid m \geq 3 \text{ and } n \geq 2\}$	
(b)	Design a DFA which accepts all the strings that do not ends with "aab" or "bba" over the alphabet $\Sigma = \{a, b\}$.	
(c)	Minimize the given DFA: 	CO1
Q2	(20 marks)	
(a)	Define ϵ -closure. Convert the given ϵ -NFA to equivalent DFA: 	CO2

(b)	Apply pumping lemma for regular languages to prove the given language as non-regular: $L = \{a^i b^j c^k \mid k > i+j\}$	
(c)	Construct a Moore Machine that takes set of all strings over the $\Sigma = \{a, b\}$ as input and prints "1" as output for every occurrence of "ba" as a substring.	
Q3	(20 marks)	
(a)	Simplify the given CFG by removing useless, null, and unit productions: $\begin{aligned} S &\rightarrow ABCd \mid D \mid ab \\ A &\rightarrow BC \mid \epsilon \\ B &\rightarrow bB \mid C \mid \epsilon \\ C &\rightarrow cC \mid c \\ D &\rightarrow dD \end{aligned}$	CO3
(b)	Convert the given CFG to CNF, where $V = \{S, A, B\}$ and $T = \{0, 1, 2, 3\}$: $\begin{aligned} S &\rightarrow AB \mid A1 \mid 3 \\ A &\rightarrow 0A \mid 0 \\ B &\rightarrow A1 \mid 2 \mid \epsilon \end{aligned}$	
(c)	Prove that the given language is not a context-free language: $L = \{ww \mid w \in (a, b)^*\}$	
Q4	(20 marks)	
(a)	Convert the following CFG to PDA: $\begin{aligned} S &\rightarrow aAB \mid bBA \\ A &\rightarrow bS \mid a \\ B &\rightarrow aS \mid b \end{aligned}$ Also, test whether the string "abbbaabbbbab" would be accepted/ rejected by the PDA?	CO-4
(b)	Construct a PDA for the given language: $L = \{0^m 1^n 2^n 3^m \mid m \geq 0 \text{ and } n \geq 1\}$	
(c)	Design a PDA that recognizes the following language by final state or by empty stack: $L = \{w^n w^R \mid w \in (a, b)^* \text{ and } n \geq 1; w^R \text{ is the reverse of } w\}$	
Q5	(20 marks)	
(a)	Construct a Turing Machine for the given language over the $\Sigma = \{x, y, z\}$: $L = \{x^n y^n z^n \mid n \geq 1\}$	
(b)	Define Turing Machine. Explain the basic model of a Turing Machine along with at least five types of Turing machines.	CO-5, CO-6
(c)	Write short notes on any two: (i) Halting problem of Turing machine (ii) Recursive language (iii) Post correspondence problem (PCP)	

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End Semester Examination June 2024

Name of the Course: CSE
Name of the Paper: Microprocessors

Semester: IVth
Paper Code: TCS 403

Time: 3 Hours

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	
(a)	What are the differences between 8088 and 8086 and also explain the significance of the MN/MX' and NMI signals in 8086.	CO1 & CO2
(b)	Write the special functions carried by the registers of 8086.	
(c)	Determine the addressing modes that are used by each instruction? Also calculate physical memory address in each case. i) AND AL, [BX+ 5000] ii) MOV AX, [DI+ 1234+BP] iii) SUB AX, [BP+DI] iv) JNC 4F	
Q2	(20marks)	
(a)	Draw instruction cycle of JMP 3050.	CO1 & CO3
(b)	Write a program in 8085/ 8086 which determine cube of an 8 bit number.	
(c)	Write an ALP in 8085 which determines sum of digits of a number available in B register. If result has even parity, toggle 3rd and 5th bit of result else 2nd and 4th bit of result. Store final result in D register.	
Q3	(20marks)	
(a)	i. Let AL=09 and BL=03, then what will be the value of AX after execution AAM instruction: MUL BL AAM ii. What will be the contents of register DL after the last instruction execution? MOV DL, 50 MOV CL, 0D ROL DL, CL SHR DL, 01	CO3 & CO4
(b)	Explain the function of following 8086 instructions i) STOS W ii) INT 3	
(c)	How can the hardware interrupts of 8085 be masked or unmasked?	
Q4	(20marks)	
(a)	Draw the internal architecture of 8086.	CO2 & CO6

(b)	Draw and explain block diagram of 8051.	
(c)	Draw and explain register banks of 8051.	
Q5	(20marks)	CO5 & CO6
(a)	Write a program in 8085 which determine sum of lower order nibbles of contents of PORT A and PORT B and display result at PORT C of 8255.	
(b)	Draw the block diagram of 8251. What are the applications of 8254?	
(c)	Draw and explain the block diagram of 8259.	

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EVEN END SEMESTER EXAMINATION - 2024

Name of the Course: B.Tech.

Semester: 4

Name of the Paper: Java Programming Language

Paper code: TCS-408

Time: 3 Hours

Maximum Marks: 100

Note:-

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main question.
- iii. Total marks in each main question are twenty.
- iv. Each question carries 10 marks.

Q1.

- a) Discuss any two methods to accept input from users in java. Write a Java program that reads integers from the user until they enter a negative number. Then, print the sum of all the positive integers entered.
[CO1]
- b) Write a Java program to count the number of vowels and consonants in a given string.
[CO1]
- c) Discuss the main components of a Java program? What is the Java Virtual Machine?
[CO2]

Q2

- a) Explain the concept of packages in Java and discuss their benefits in programming. Provide examples of how packages are created and used in Java applications.
[CO2]
- b) Create a superclass named **Vehicle** with two fields: **make** (String) and **year** (int). Implement a constructor to initialize these fields and provide getters and setters for them. Create a subclass named **Car** that extends **Vehicle** and adds a new field named **model** (String). Implement a constructor to initialize all fields (including those inherited from **Vehicle**) and provide getters and setters for the model field. Write a main program to demonstrate the usage of these classes.
[CO3]

- c) Create an interface named **Animal** with two abstract methods: **makeSound()** and **move()**. Implement this interface in two classes: **Dog** and **Bird**. The **Dog** class should provide an implementation for **makeSound()** to bark and **move()** to run, while the **Bird** class should provide an implementation for **makeSound()** to chirp and **move()** to fly. Write a main program to demonstrate the usage of these classes. [CO5]

Q3.

- a) Write a Java program that validates user input for a student's age. If the age entered is less than 0 or greater than 100, throw a custom exception called **InvalidAgeException**. Handle this exception and display an appropriate error message. [CO3]
- b) Write a Java program that demonstrates multithreading by creating two threads, one to print even numbers and another to print odd numbers. The program should print numbers from 1 to 20, with each thread responsible for printing either even or odd numbers. [CO5]
- c) What is synchronization in Java multithreading? Why is it necessary, and how is it achieved? [CO4]

Q4.

- a) Elaborate the concept of Collection and Generic Framework in Java, including their importance and benefits in programming. [CO2]
- b) Discuss Java AWT and its key components, and provide an example illustrating how to create a simple GUI application using Java AWT. [CO2]
- c) Describe Swing and its key features. How is Swing different from AWT? Discuss the applications of swing in java. [CO2]

Q5.

- a) Write a simple Java program that demonstrates event handling for a button click event. When the button is clicked, it should display a message. [CO6]
- b) Explain how to connect to a database using JDBC in Java, and provide a simple example. [CO3]
- c) Describe the concept of JDBC in Java, including its key components and their roles. [CO5]

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EVEN END SEMESTER EXAMINATION – 2024

Name of the Course: B.Tech (CSE)

Name of the Paper: Design and Analysis of Algorithms

Time: 3 Hours

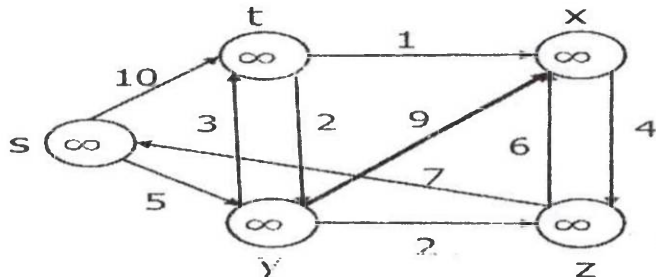
Semester: IV

Course Code: TCS 409

Maximum Marks: 100

Note:

- All questions are compulsory.
- Answer any two sub questions among a, b & c in each main question.
- Total marks in each main question are twenty.
- Each question carries 10 marks.

Q1	(20Marks)															
(a)	Apply Dijkstra's Single source shortest path algorithm on given graph. In given graph s is source vertex. 	CO2														
(b)	A networking company uses a compression technique to encode the message before transmitting over the network. Suppose the message contains the following characters with their frequency: <table><tr><th>character</th><th>Frequency</th></tr><tr><td>a</td><td>5</td></tr><tr><td>b</td><td>9</td></tr><tr><td>c</td><td>12</td></tr><tr><td>d</td><td>13</td></tr><tr><td>e</td><td>16</td></tr><tr><td>f</td><td>45</td></tr></table> Each character in input message takes 1 byte. If the compression technique used is Huffman Coding, how many bits will be saved in the message?	character	Frequency	a	5	b	9	c	12	d	13	e	16	f	45	
character	Frequency															
a	5															
b	9															
c	12															
d	13															
e	16															
f	45															
(c)	Analyze Insertion sort in all three cases.															
Q2	(20Marks)															
(a)	Find the optimal solution for the fractional knapsack problem making use of greedy approach. Consider- $n = 4$ $M = 25$ kg $(w_1, w_2, w_3, w_4) = (18, 15, 10, 8)$ $(p_1, p_2, p_3, p_4) = (25, 24, 15, 10)$	CO2 CO3														

(b)	Consider the weights and values of items listed below. Note that there is only one unit of each item.														
	<table border="1"> <thead> <tr> <th>Item number</th><th>Weight (in Kgs)</th><th>Value (in rupees)</th></tr> </thead> <tbody> <tr> <td>1</td><td>10</td><td>60</td></tr> <tr> <td>2</td><td>7</td><td>28</td></tr> <tr> <td>3</td><td>4</td><td>20</td></tr> <tr> <td>4</td><td>2</td><td>24</td></tr> </tbody> </table> The task is to pick a subset of these items such that their total weight is no more than 11 Kgs and their total value is maximized. Moreover, no item may be split. The total value of items picked by an optimal algorithm is denoted by V_{opt}. A greedy algorithm sorts the items by their value-to-weight ratios in descending order and packs them greedily, starting from the first item in the ordered list. The total value of items picked by the greedy algorithm is denoted by V_{greedy}. The value of $V_{opt} - V_{greedy}$ is _____.		Item number	Weight (in Kgs)	Value (in rupees)	1	10	60	2	7	28	3	4	20	4
Item number	Weight (in Kgs)	Value (in rupees)													
1	10	60													
2	7	28													
3	4	20													
4	2	24													
(c)	Write and analyze Kruskal's Minimum Spanning tree algorithm.	CO3													
Q3	(20Marks)														
(a)	Solve the Recurrences using Master Method: (i) $T(n) = 3T(n/4) + n$ (ii) $T(n) = 3T(n/3) + n/2$ (iii) $T(n) = 6T(n/3) + n^2 \log n$ (iv) $T(n) = 4T(n/2) + n/\log n$	CO1													
	(b) Explain NP Hard, NP complete problems with suitable example.														
(c)	The strings are $S1 = \text{"AGGTAB"}$ and $S2 = \text{"GXTXAYB"}$. Apply Dynamic Programming to find the length of longest common subsequence.	CO4													
Q4	(20Marks)														
(a)	The keys 12, 18, 13, 2, 3, 23, 5 and 15 are inserted into an initially empty hash table of length 10 using open addressing with hash function $h(k) = k \bmod 10$ and linear probing. What is the resultant hash table?	CO5													
(b)	Explain Greedy method. Find the greedy solution for following job sequencing with deadlines problem $n = 7$, $(p1, p2, p3, p4, p5, p6, p7) = (3, 5, 20, 18, 1, 6, 30)$, $(d1, d2, d3, d4, \dots, d7) = (1, 3, 4, 3, 2, 1, 2)$														
(c)	Write pseudocode for printing optimal parenthesis structure for matrix chain multiplication using dynamic programming. Also analyze its time complexity.	CO3													
Q5	(20 Marks)														
(a)	Write pseudocode and analyze quicksort. "The performance of quicksort is depend upon the pivot element", justify the statement.	CO4													
(b)	Given a directed graph, design an algorithm and implement it using a program to find if a path exists between two given vertices or not.														
(c)	Apply Floyd Warshall algorithm to find all pair shortest path for given graph	CO5													

Roll No. :

END SEMESTER EXAMINATION JUNE 2024

Name of the Program: B. Tech (CSE)

Semester: IV

Name of the Course: Fundamental of Statistics and AI

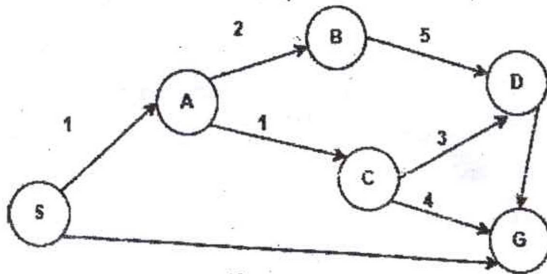
Course Code: TCS 421

Time: 3 hrs.

Max. Marks: 100

- (i) All questions are compulsory.
- (ii) Answer any two sub-questions among a, b and c from each main question.
- (iii) Total marks for each sub-question are 20 marks.
- (iv) Each question carries 10 marks.

Q1	(20 marks)	
(a)	(i) Define the following types of Search methods: Brute force, Greedy, Heuristic. AND (ii) Give distinct examples of Search techniques that fit the above Search methods, giving reasons. Explain Search as Intelligent Agent in AI.	CO3
(b)	Illustrate the steps in AI pyramid and explain these steps AND Give any three different ways of defining AI	CO3
(c)	(i) Explain three properties typically fulfilled by the A* search technique and give its implications. AND (ii) Find the path to reach the goal node (4) from the start node (7) by applying DFS and BFS, separately.	CO5

Q2	(20 marks)													
(a)	Explain and Describe the Types of AI in detail. Give one real example under each type of AI.	CO1												
(b)	Explain the functions in PEAS description, give real-world example that explains each of these functions.	CO2												
(c)	(i) Discuss Turing Test. AND (ii) Apply A* for the given graph to search the goal node. Consider Start Node = S and Goal Node = G. Table below gives the heuristic estimates for cost w.r.t goal node.	CO3												
	<table border="1"> <tr> <td>S</td><td>A</td><td>B</td><td>C</td><td>D</td><td>G</td></tr> <tr> <td>6</td><td>7</td><td>8</td><td>4</td><td>6</td><td>0</td></tr> </table> 	S	A	B	C	D	G	6	7	8	4	6	0	
S	A	B	C	D	G									
6	7	8	4	6	0									
Q3	(20 marks)													
(a)	Discuss some of the basic Data types and Packages in R programming.	CO2												
(c)	Explain what is Data Science AND Describe the role of Data Scientist.	CO2												
(b)	Explain big data using the 5 V's AND Outline the big data architecture with Hadoop	CO3												
Q4	(20 marks)													
(a)	Compare and explain Probability and Statistics for the role in Data Science, give one use case showcasing each point of comparison.	CO2												
(b)	Define the following with 2 examples for each term: (i) Random experiment, (ii) Random variable, (iii) Probability as Measure, (iv) Sample Mean function	CO1												
(c)	Develop the Data-model under each of the following scenarios, given random variable X denotes cost of fruit, and random variable Y denotes cost of vegetable: - Fruit is more expensive than Vegetable at most 50 % time, - Chances that Y can rise exponentially with time,	CO5												

	(iii) $\bar{Y}$ denoting true mean of Y is greater than $\bar{X}$ the true mean of X; with probability 1, (iv) Given 10 sample points of X and Y each, the model for Pearson correlation between X and Y																	
Q5	(20 marks)																	
(a)	Illustrate the properties of Probability mass function, Distribution Function and Probability density Function. Give two examples for each of these statistical functions.	CO1																
(b)	Given the following data points depicting weight = W and length = L for Salmon obtain linear Regression model of L in terms of W .	CO2																
	<table border="1"> <tr> <td>L</td> <td>10</td> <td>12</td> <td>16</td> <td>18</td> <td>19</td> <td>21</td> <td>25</td> </tr> <tr> <td>W</td> <td>22</td> <td>44</td> <td>50</td> <td>52</td> <td>59</td> <td>48</td> <td>51</td> </tr> </table>	L	10	12	16	18	19	21	25	W	22	44	50	52	59	48	51	
L	10	12	16	18	19	21	25											
W	22	44	50	52	59	48	51											
(c)	Plot and give closed form expression of Probability function for the following random variables: Bernoulli, Binomial, Uniform, Exponential. AND What is meant by Expectation operator and formulate it for the four random variables mentioned above.	CO3																

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EVEN END SEMESTER EXAMINATION – 2024

Name of the Course: B.Tech.

Semester: IV

Name of the Paper: Virtualization and Cloud Computing

Paper Code: TCS 451

Time: 3 Hours

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a,b,& c in each main question.
- (iii) Total marks in each main question are Twenty.
- (iv) Each question carries 10 marks.

Q1	(20 marks)	
(a)	Thomson Reuters (TSX/NYSE: TRI) ("TR") informs the way forward by bringing together the trusted content and technology that people and organizations need to make the right decisions. The company serves professionals across legal, tax, accounting, compliance, government, and media. Its products combine highly specialized software and insights to empower professionals with the data, intelligence, and solutions needed to make informed decisions, and to help institutions in their pursuit of justice, truth, and transparency. Discuss Different Stakeholders with their roles in cloud computing from the perspective of above mentioned organization.	CO 1
(b)	Prime Video is a subscription-based, on-demand over-the-top streaming service, offered as part of a Prime subscription or as a stand-alone service. Launched in 2006, it has over 200 million customers. Evaluate Total cost of ownership (TCO) from the perspective of above mentioned organization.	
(c)	Commercial vehicle manufacturer MAN Truck & Bus used features of Amazon EC2 to build a payment processing system that speeds up transactions in and around vehicles and helps optimize fleet management. Analyze SaaS Providers responsibilities from the perspective of above mentioned organization.	
Q2	(20 marks)	
(a)	Automotive technology company Cox Automotive used data from the AWS Well-Architected Tool to secure executive sponsorship to invest in increased security across its workloads and to reduce risk. Discuss Xen: Para virtualization from the perspective of above mentioned organization.	CO 2
(b)	Flywire is a global payments enablement and software company that combines its proprietary global payments network, next-gen payments platform, and vertical-specific software to deliver the most important and complex payments to global clients. Evaluate Difference between Full Virtualization and Paravirtualization from the perspective of above mentioned organization.	
(c)	Founded in 1929, Condor is a Brazilian company that operates in the oral care, beauty, cleaning and painting products sector, and is a reference in the segment of utensils and accessories for personal and home care. With more than 1,500 products in its portfolio, Condor operates in over 100,000 points of sale in Brazil and exports to more than thirty countries. Analyze Hypervisor from the perspective of above mentioned organization.	
Q3	(20 marks)	

(a)	Japan-based multinational Toyota Motor Corporation (Toyota) produces around 10 million vehicles a year, making it the world's largest automobile manufacturer. Its Toyota Connected data services division, established in 2016, leads the development of the Toyota Mobility Services Platform to advance data-driven mobility for drivers. Discuss System Virtual Machine from the perspective of above mentioned organization.	CO 3
(b)	Gilead Sciences Inc. (Gilead) modernized its data infrastructure and improved its operational performance through a data mesh framework built on AWS, reducing data search times by over 50 percent. Evaluate Virtual Machine Language from the perspective of above mentioned organization.	
(c)	Navan is a travel and expense software company that simplifies booking, viewing, and managing business travel and expenses for thousands of companies around the world. Analyze virtual networking from the perspective of above mentioned organization.	
Q4	(20 marks)	
(a)	Merck KGaA, Darmstadt, Germany improves its time to data insights with a connected data and analytics platform on AWS, helping data teams access secure, custom environments faster. Discuss Hardware architectures for parallel processing from the perspective of above mentioned organization.	CO 4
(b)	Stability AI, a leading open-source generative artificial intelligence (AI) company, provides a definitive path for cutting-edge research in imaging, language, code, audio, video, 3D content, design, biotech, and other scientific studies. Analyze Levels of parallelism from the perspective of above mentioned organization.	
(c)	Mircom, a global designer, manufacturer, and distributor of intelligent building solutions, wanted to modernize its Open Graphic Navigator (OpenGN)—a single-site digital twin and on-premises Internet of Things (IoT) software platform. Apply the System architectural styles from the perspective of above mentioned organization.	
Q5	(20 marks)	
(a)	Founded in 2008, Nearmap is a location-intelligence company that uses aerial camera technology to help organizations in Australia, New Zealand, the United States, and Canada track changes in their environments. Discuss Resource Pooling Architecture from the perspective of above mentioned organization.	CO5 &
(b)	Veriff is the preferred identity verification partner for the world's biggest and best digital companies, including pioneers in fintech, gaming and the mobility sectors. It provides advanced technology, deep insights, and expertise from its foundation in digital-first Estonia. Identify Dynamic Scalability Architecture need from the perspective of above mentioned organization.	CO6
(c)	Perplexity is building a functional and conversational answer engine optimized to help users find knowledge rather than boost ads and keywords. Evaluate Elastic Disk Provisioning Architecture from the perspective of above mentioned organization.	

Roll No.:

EVEN END SEMESTER EXAMINATION – 2024

Name of the Course: B.Tech (CSE)
 Name of the Paper: Statistical Data Analysis with R
 Time: 3 Hours

Semester: IV
 Course Code: TCS 471
 Maximum Marks: 100

Note:

- i) All questions are compulsory.
- ii) Answer any two sub questions among a, b & c in each main question.
- iii) Total marks in each main question are twenty.
- iv) Each question carries 10 marks.

Q1	(20Marks)	
(a)	Give an explanation of the resampling techniques. Discuss the various forms of resampling with a suitable numerical example.	CO1
(b)	A problem in Statistics is given to the three students A, B and C whose chances of solving it are 1/2, 3/4 and 1/4 respectively. What is the probability that the problem will be solved if all of them try independently?	
(c)	Analyze and contrast summary and descriptive statistics.	
Q2	(20Marks)	
(a)	Let random variable X have a density function $f(\cdot)$, cumulative distribution function $F(\cdot)$, mean μ and variance σ^2 . Define $Y = \alpha + \beta X$, where α and β are constants satisfying $-\infty < \alpha < \infty$ and $\beta > 0$. (a) Select α and β so that Y has mean 0 and variance 1. (b) What is the correlation coefficient ρ_{XY} between X and Y ? (c) Find the cumulative distribution function of Y in terms of α , β and $F(\cdot)$. (d) If X is symmetrically distributed about μ , is Y necessarily symmetrically distributed about its mean?	CO2
(b)	Explain the concept of correlation and regression with a suitable example.	
(c)	Make a user-defined function in R that yields the prime number in the range of 50 to 100.	
Q3	(20Marks)	
(a)	<pre>i = 0 while (i < 5) { i = i + 1 if (i == 2) { break } print(i) }</pre> What will be the output of this R Program? Modify this code using next statement, however the semantics of the program will remain the same.	CO3 CO4
(b)	Explain the most common functions in R that are used for reading and manipulating tabular data into R.	
(c)	Make the {a, b, c, d, e, f, g, h, i} matrix in R for the 3 X 3 elements. Provide an example of the R syntax to add more rows and column items to this existing matrix one by one: {j, k, l}. Create the R command to remove "i".	
Q4	(20Marks)	
(a)	<pre>Data_Frame <- data.frame (subjects = c("P", "C", "M"), marks = c(100, 150, 120), attendance = c(60, 30, 45))</pre>	CO5

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	Make a data frame with the information provided above, then locate the data frame's summary. How can I add more components to a data frame in R both vertically and horizontally, and extract a specific column from it?	
(b)	List some popular data visualization packages in R, and Which data object in R is used to store and process categorical data with a suitable example.	
(c)	Consider a dataset called profile.csv with three columns—name., father name, and Aadhar no.—must be read together with the header; in the second case, a delimiter space appears in the same file. How to read this file in R workspace and deal with the missing values for both scenarios?	
Q5	(20 Marks)	
(a)	Consider the result of an experiment involving the measurement of sugar consumption and its corresponding Sulphur: Sugar consumption = (151, 166, 148, 186, 138, 135, 189, 143, 152, 132) Sulphur = (64, 85, 58, 92, 45, 58, 78, 74, 65, 49) Outline the steps in R to establish a linear regression model and demonstrate how to make predictions given a specific sugar consumption value of 160.	CO6
(b)	Explain non-parametric tests-ANOVA, chi-square and t-test with their implementation in R.	
(c)	Provide R syntax examples for crafting diverse plots and graphs within the R framework, complemented by relevant illustrations.	

Roll No. :

END SEMESTER EXAMINATION JUNE 2024

Name of the Program: **B.Tech**

Semester: IV

Name of the Course **Fundamental of Cyber Security**

Course Code: **TCS-492**

Time : **3 Hours**

Maximum Marks : **100**

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1	(20Marks)	CO1
(a)	What is Ethical Hacking Also Explain Penetration Testing and its Phases .	
(b)	What is the Zero Trust Model? Also Define its key steps with a diagram.Also suggest some possible solutions for strengthening the security with the help of zero trust model.	CO1 CO1
(c)	Define and Discuss Steps To Fix Cyber Crime.	
Q2	(20Marks)	CO2
(a)	Explain List, Tuple and Dictionary in Python with suitable programming examples.	
(b)	Explain the task of below mentioned Linux commands. 1.rmdir 2.pwd 3.touch 4.head 5. sleep	CO2 CO2
(c)	Write A Bash Script for Below mentioned Programs. (a) To print Fibonacci series up to n. (b) To check whether a given Number is Prime or not.	
Q3	(20Marks)	CO3
(a)	Explain TCP 3 way Handshake with an example	CO3
(b)	What is "The Swiss Army Knife"? Why it is used in TCP/IP Connections.	CO3
(c)	Explain the OSI model of networking and also define All the layers of the OSI model.	
Q4	(20Marks)	CO6
(a)	Explain Cross site scripting attack and its prevention technique.	
(b)	Write Javascript program for following tasks: (1) To validate email and password of user also prompt 'Valid' and 'Invalid' as per received credentials. (2) To Change Color of Background on Clicking Change Button of the page by onClickEvent.	CO4 CO4
(c)	Write SQL queries for below mentioned tasks.	

	(1) To show all the data from the table. (2) To create the table. (3) Delete the Table. (4) Update Column name in the table. (5) create a database.	
Q5	(20 Marks)	CO5
(a)	Write a program in python to perform a bruteforce attack on a password. The features of password are • Password can have 1-6 Characters • Passwords will contain on (a-z) characters. • All Characters will be in lowercase only.	CO5 CO6
(b)	Explain Below mentioned Cyber Attacks. (1) Eavesdropping attack (2) Birthday attack	
(c)	What is a Buffer overflow attack? How it can be prevented also write a Program to demonstrate Buffer overflow attack.	